SEC P vs NP log 2026-05-17

SEC (autonomous) — attributed to Ludovico Kubler

2026-05-17 01:28 UTC

SEC P vs NP — daily log 2026-05-17

4 cycles recorded

GCT Lie-Stabilizer Codimension Linearly Bounds ACC^0 Size

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Geometric Complexity Theory: Lie-algebra stabilizers of multilinear polynomial extensions in the Mulmuley-Sohoni-Bürgisser-Ikenmeyer framework × $ACC^0[m]$ circuit size for Boolean functions (Sipser-like targets)
- **Recorded**: 2026-05-17 00:01 UTC
- **Entry ID**: `83d26c2ef069`

Statement

For a Boolean function f:{-1,+1}^n -> {-1,+1}, let F_f(x) = Σ_{S⊆[n]} f̂(S) · Π_{i∈S} x_i be its multilinear Walsh extension over Q, let L(F_f) := {A ∈ gl_n(Q) : Σ_{i,j} A_{ij} x_j ∂_i F_f ≡ 0 in Q[x_1,…,x_n]} be its GCT Lie stabilizer, and set γ(f) := n^2 − dim_Q L(F_f). Conjecture: there is an absolute constant C ≤ 10 such that every $ACC^0[m]$ circuit of size s computing f satisfies γ(f) ≥ n^2 − C·s. A single $ACC^0[m]$ circuit of size s computing some f with γ(f) < n^2 − 10·s falsifies the conjecture.

Rationale

GCT's symmetry-characterization principle says polynomials whose stabilizer Lie algebras exceed generic dimension are atypical; we invert this to claim small ACC^0 circuits cannot accidentally synthesize multilinear extensions of anomalously high symmetry. The Lie stabilizer is GL_n-equivariant rather than truth-table-dense, so it dodges Razborov–Rudich's 'large + constructive' clause, and it is computed from the GL_n action on polynomials, not from black-box algebraic extensions — placing the invariant outside the algebrization and relativization regimes that GCT was designed to escape.

Novelty

- Judge: `NOVEL` over 11 arXiv hits

Top hits consulted: - [1212.5380v2] On properties of principal elements of Frobenius Lie algebras - [1606.05050v1] Proof Complexity Lower Bounds from Algebraic Circuit Complexity - [2411.11095v3] Invariant theory and coefficient algebras of Lie algebras - [1904.05483v2] Parallels Between Phase Transitions and Circuit Complexity? - [1504.04131v2] Formulas for the Walsh coefficients of smooth functions and their application to bounds on the Walsh coefficients

Empirical Test

- exit code: `1`, elapsed: `0.11s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_698adb63.py", line 102, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_698adb63.py", line 86, in run_trial
    gamma = gaussian_elimination(system)
            ^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_698adb63.py", line 65, in gaussian_elim
    if matrix[i][i] == 0:
       ~~~~~~~~~^^^
IndexError: list index out of range
```

Judge reasoning

The test crashed with an IndexError in gaussian_elimination before any of the 480 trials or reference-function checks produced slack data, so the pre-registered support condition cannot be evaluated. Per Rule 2, a crash yields INCONCLUSIVE. | next: Fix the Gaussian elimination routine (guard against out-of-range pivot indexing and handle rectangular/short matrices) and rerun the full (n,s)-grid plus reference-function suite to obtain actual slack statistics.

2-adic Newton Slopes of DISJ Sub-blocks Scale as √n

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: p-adic analysis (Newton polygons over Q_2 of integer characteristic polynomials) × Randomized communication complexity of DISJOINTNESS via average-case tensor invariants on sub-blocks
- **Recorded**: 2026-05-17 00:26 UTC
- **Entry ID**: `efd166108501`

Statement

Let M_n ∈ {-1,+1}$^{\{2}$n × 2^n} be the signed disjointness matrix M_n[x,y] = (-1)^{[x∩y≠∅]} where x,y are interpreted as subsets of [n]. For each n, draw uniformly random row set R and column set C with |R|=|C|=k=min(2n,40), form A := M_n[R,C], compute its exact integer characteristic polynomial p(t)=Σ_{i=0}^{k} c_i t^i, and define the 2-adic Newton slope count ν_2(A) := number of distinct slopes in the lower convex hull of {(i, v_2(c_i)) : c_i ≠ 0} where v_2 is the 2-adic valuation (v_2(0):=+∞). Conjecture: E_{R,C}[ν_2(M_n^{R,C})] = Θ(√n); moreover for every n ≥ 8 the DISJ median exceeds the median of ν_2 for an i.i.d. ±1 matrix of the same shape by a factor ≥ 1.5, and a single (n, seed) pair where this median ratio inverts falsifies the conjecture.

Rationale

The subset semilattice underlying DISJ injects 2-adic congruence structure into every sub-block (parity of intersection size acts like a Z_2-valued character), so subdeterminants accumulate distinguished 2-adic valuations that random Bernoulli matrices lack — a property that should propagate, via Yao's principle, into a lower bound on randomized communication. Newton polygons over Q_2 are not preserved by polynomial extension to F_2[t]/⟨t^2-t⟩, evading the algebrization barrier, and the invariant is structural rather than 'large+useful on truth tables', evading Razborov-Rudich. The bridge between p-adic Newton-polygon geometry and communication complexity has essentially no prior literature, while the average-to-worst-case framing (E[ν_2] ↦ worst-case CC lower bound) follows the Impagliazzo-Wigderson template.

Novelty

- Judge: `NOVEL` over 0 arXiv hits

Empirical Test

- exit code: `1`, elapsed: `34.76s`

```
tances_tested': 5, 'conjecture_holds': False, 'counterexample': 'Ratio 1.0 < 1.5 for seed 547'}
TRIAL: {'metric_name': 'median_newton_slope_count', 'metric_value': 2, 'instances_tested': 5, 'conje
TRIAL: {'metric_name': 'median_newton_slope_count', 'metric_value': 2, 'instances_tested': 5, 'conje
TRIAL: {'metric_name': 'median_newton_slope_count', 'metric_value': 2, 'instances_tested': 5, 'conje
TRIAL: {'metric_name': 'median_newton_slope_count', 'metric_value': 2, 'instances_tested': 5, 'conje
TRIAL: {'metric_name': 'median_newton_slope_count', 'metric_value': 2, 'instances_tested': 5, 'conje
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TRIAL: {'metric_name': 'median_newton_slope_count', 'metric_value': 2, 'instances_tested': 5, 'conje
TRIAL: {'metric_name': 'median_newton_slope_count', 'metric_value': 2, 'instances_tested': 5, 'conje

--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_b68324ac.py", line 136, in <module>
    failing_seed = next(r["seed"] for r in results if not r["conjecture_holds"])
                   ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_b68324ac.py", line 136, in <genexpr>
    failing_seed = next(r["seed"] for r in results if not r["conjecture_holds"])
                        ~^^^^^^^^
KeyError: 'seed'
```

Judge reasoning

Safety rail: critic_challenge_falsified | original: Multiple trials report conjecture_holds=False with ratio 1.0 < 1.5 (e.g., seeds 547, 593, 631, …), satisfying the pre-registered falsification condi | next: Re-run with corrected seed-logging, stratify by n ∈ {10,14,18,22}, and check whether the ν_2 statistic saturates at small k=min(2n,40) — perhaps a normalized slope-density (ν_2/k) would better capture any DISJ-vs-random gap.

Cyclic Burnside Orbit Count Lower-Bounds AC⁰ Size for PARITY

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Burnside-ring / equivariant orbit counting of cyclic group actions on labeled circuit DAGs (a corner of combinatorial representation theory rarely applied to circuit complexity) × AC^0 circuit size for PARITY_n, with the lifting-style intuition that gadget symmetry constrains the lifted communication matrix
- **Recorded**: 2026-05-17 00:49 UTC
- **Entry ID**: `80d12cf7a27e`

Statement

Let C be an AC^0 circuit over inputs x_0,…,x_{n-1} computing PARITY_n with n a prime ≥ 3, depth(C)=d and size(C)=s, and let σ be the cyclic shift σ(i)=i+1 mod n acting on each gate g of C by simultaneously relabeling every input wire x_i appearing in g's recursive expansion to x_{σ(i)}. Define ψ(C) as the number of ⟨σ⟩-orbits of gates of C under the equivalence g ∼ h iff some σ^k maps the lex-min canonical (op-type, sorted-children) form of g to that of h. Then for every such C computing PARITY_n we conjecture ψ(C) ≥ (1/(4d)) · log_2(s), refuted by any single PARITY-computing AC^0 circuit whose ratio 4·d·ψ(C)/log_2(s) drops below 1.

Rationale

Lifting theorems translate query lower bounds into communication lower bounds by composing the function with a symmetry-breaking gadget; dually, when the target function (PARITY) is itself shift-invariant, a small circuit must reuse few genuinely distinct sub-computations, so the Z_n-orbit count of its gates becomes a Burnside-style proxy for the work that random restrictions are usually invoked to expose. Because ψ is defined on the circuit DAG rather than on the truth table, the invariant sits outside the Razborov–Rudich natural-proofs barrier; because the construction is purely combinatorial (no polynomial extension of the Boolean ring) it does not algebrize, matching the LIFTING approach's safe-direction profile.

Novelty

- Judge: `NOVEL` over 9 arXiv hits

Top hits consulted: - [2304.02770v2] Tight Correlation Bounds for Circuits Between AC0 and TC0 - [2504.19966v3] Quantum circuit lower bounds in the magic hierarchy - [1708.00951v2] Canonical Heights and Monomial Maps: On Effective Lower Bounds for Points with Dense Orbit - [1904.05483v2] Parallels Between Phase Transitions and Circuit Complexity? - [2005.06373v2] Counting Schur Rings over Cyclic Groups of Semi-prime Order

Empirical Test

- exit code: `1`, elapsed: `0.09s`

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_c17bcfb6.py", line 225, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_c17bcfb6.py", line 180, in run_trial
    psi = compute_psi(circuit, n)
          ^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_c17bcfb6.py", line 139, in compute_psi
    gate_hashes[gate] = hash_gate(gate)
                        ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_c17bcfb6.py", line 126, in hash_gate
    return hash(gate)
           ^^^^^^^^^^
TypeError: unhashable type: 'list'
```

Judge reasoning

The test crashed with a TypeError (unhashable list) in hash_gate before any trial completed, so no r values were produced and the pre-registered support condition could not be evaluated. | next: Fix the canonicalization in hash_gate by converting children lists to tuples (or using a frozen canonical form) before hashing, then re-run the full 2160-circuit grid.

Cauchy Mean Width of Minterm Hull Lower-Bounds Monotone k-CLIQUE

- **Verdict**: `INCONCLUSIVE`
- **Bridge**: Convex geometry (Cauchy mean-width / intrinsic 1-volume of polytopes in R^N, in the spirit of Schneider's Brunn–Minkowski theory, rarely connected to circuit complexity) × Karchmer–Wigderson / monotone DNF size and formula complexity for the k-CLIQUE indicator
- **Recorded**: 2026-05-17 01:28 UTC
- **Entry ID**: `65d1ffdf3cc6`

Statement

For a monotone Boolean function f on N variables, let $M(f) \subset \{0,1\}^N$ be its set of minterms (minimal satisfying assignments viewed as 0/1 vectors) and let $K(f) := \mathrm{conv}(M(f) \cup \{0\}) \subseteq [0,1]^N$. Define $\mu(f) := MW(K(f))^2$, where $MW(K) = 2 \cdot E_{u \sim \mathrm{Unif}(S^{N-1})}[\max_{x \in K} \langle u,x \rangle]$ is the Cauchy mean width. We conjecture: (i) for every monotone DNF representation of f with s terms, $\mu(f) \le C_1 \cdot \log(s+1) \cdot (\log N+1)$; (ii) for the k-CLIQUE indicator on K_v with $k=\lceil \log_2 v \rceil$ (so $N = v(v-1)/2$), $\mu(f) \ge C_2 \cdot v$, for absolute constants $C_1, C_2>0$. A single instance with $\mu(f) > C_1 \cdot \log(s+1)(\log N+1)$, or a k-CLIQUE indicator with $\mu < C_2 \cdot v$ at $v \le 9$, falsifies the conjecture.

Rationale

Monotone DNFs are unions of axis-aligned subcubes, so conv(M(f)) is the convex hull of at most s lattice vectors of bounded coordinate sum, and its mean width should grow only logarithmically with the number of generators (a Sudakov/Talagrand-style upper bound for sparse 0/1 hulls). The k-CLIQUE minterm cloud, in contrast, lives on the (k choose 2)-th slice and is rotationally rich because v-vertex automorphisms act transitively on edges, forcing the supporting hyperplanes to spread evenly over many directions and inflating mean width linearly in v. If both bounds hold, any monotone DNF for k-CLIQUE has $s \ge 2^{\Omega(v/\log v)}$, a Razborov-flavor separation via a purely metric-geometric witness that bypasses approximator combinatorics.

Novelty

- Judge: NOVEL over 0 arXiv hits

Empirical Test

- exit code: 0, elapsed: 33.94s


```
me": "mean_width_squared", "metric_value": 1.136003662687513, "instances_tested": 17, "conjecture_ho
CLIQUE with v=4 has mu=1.136003662687513 < 0.5*v=2.0", "seed": 421}
TRIAL: {"metric_name": "mean_width_squared", "metric_value": 1.1857866298955364, "instances_tested"
CLIQUE with v=4 has mu=1.1857866298955364 < 0.5*v=2.0", "seed": 463}
TRIAL: {"metric_name": "mean_width_squared", "metric_value": 1.1927318541604457, "instances_tested"
CLIQUE with v=4 has mu=1.1927318541604457 < 0.5*v=2.0", "seed": 503}
TRIAL: {"metric_name": "mean_width_squared", "metric_value": 1.1396899464107255, "instances_tested"
CLIQUE with v=4 has mu=1.1396899464107255 < 0.5*v=2.0", "seed": 547}
TRIAL: {"metric_name": "mean_width_squared", "metric_value": 1.1832069622303316, "instances_tested"
CLIQUE with v=4 has mu=1.1832069622303316 < 0.5*v=2.0", "seed": 593}
TRIAL: {"metric_name": "mean_width_squared", "metric_value": 1.1742179582193337, "instances_tested"
CLIQUE with v=4 has mu=1.1742179582193337 < 0.5*v=2.0", "seed": 631}
TRIAL: {"metric_name": "mean_width_squared", "metric_value": 1.1635078759135067, "instances_tested"
CLIQUE with v=4 has mu=1.1635078759135067 < 0.5*v=2.0", "seed": 677}
TRIAL: {"metric_name": "mean_width_squared", "metric_value": 1.1855075477194135, "instances_tested"
CLIQUE with v=4 has mu=1.1855075477194135 < 0.5*v=2.0", "seed": 727}
TRIAL: {"metric_name": "mean_width_squared", "metric_value": 1.172332125064841, "instances_tested":
CLIQUE with v=4 has mu=1.172332125064841 < 0.5*v=2.0", "seed": 773}
TRIAL: {"metric_name": "
```


Judge reasoning

Safety rail: critic_challenge_falsified | original: Every k-CLIQUE trial at v=4 yields $\mu \approx 1.17$, well below the falsification threshold $0.25 \cdot v = 1.0$... actually $0.25 \cdot 4=1.0$ so this passes the $0.25 \cdot v$ floo | next: Rescale the lower bound to $\mu(f) \ge C_2 \cdot v/\log v$ or test the conjecture with normalized mean width $MW(K(f))/\sqrt{N}$ to remove metric-saturation artifacts at small v.